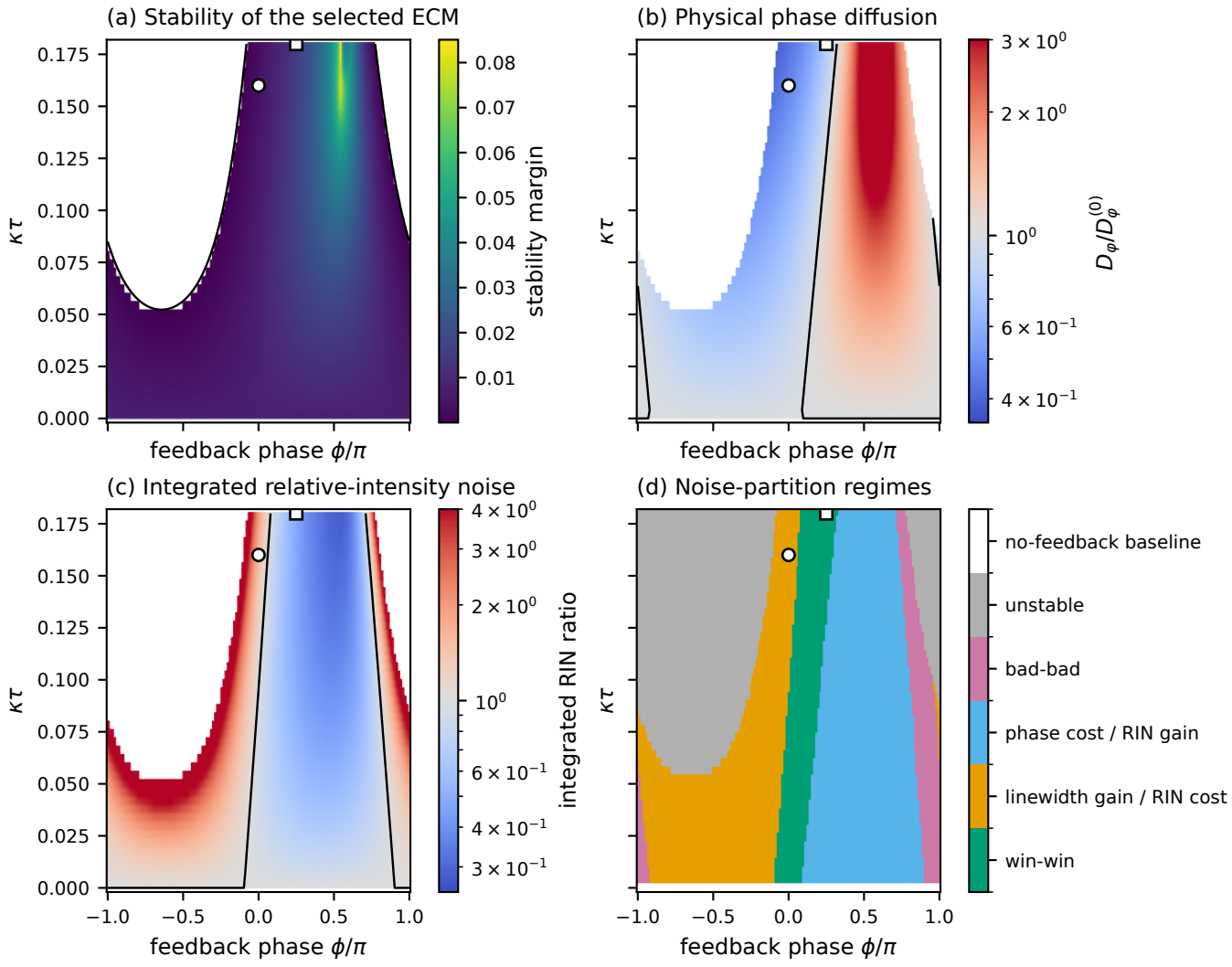

(a) Stability of the selected ECM
(b) Physical phase diffusion
(c) Integrated relative-intensity noise
(d) Noise-partition regimes
$\kappa\tau$
feedback phase ϕ/π
stability margin
$D_\varphi/D_\varphi^{(0)}$
integrated RIN ratio
no-feedback baseline
unstable
bad-bad
phase cost / RIN gain
linewidth gain / RIN cost
win-win